

Network Troubleshooting Assessment

Section A

1. Explain how Deep Packet Inspection (DPI) at the Network and Transport layers of the OSI model allows a startup to isolate specific traffic bottlenecks causing slow internet speeds despite high available bandwidth.

Answer: Traditional packet filtering only examines packet headers (like IP addresses and ports). In contrast, Deep Packet Inspection (DPI) inspects both the header and the data part (payload) of packets as they pass a checkpoint. Even though DPI typically operates up to the Application layer to identify specific protocols or apps, looking at it through the lens of Network (Layer 3) and Transport (Layer 4) properties combined with payload patterns allows a startup to:

Identify non-business traffic (e.g., video streaming or peer-to-peer file sharing) hidden over standard ports like HTTP/HTTPS (80/443).

Detect anomalous traffic spikes, packet drops, or excessive TCP retransmissions indicative of a local bottleneck or network attack (e.g., DDoS).

Isolate applications consuming excessive bandwidth, allowing administrators to apply Quality of Service (QoS) rules or rate-limiting to preserve bandwidth for critical business functions.

2. Explain the distinct roles of DHCP and DNS in a scenario where startup devices are successfully assigned IP addresses but remain unable to resolve external web addresses for internet access.

Answer: DHCP (Dynamic Host Configuration Protocol) and DNS (Domain Name System) serve separate but complementary roles:

DHCP: Its role is to dynamically assign local network configurations to client devices, including an IP address, subnet mask, default gateway, and DNS server addresses. In this scenario, DHCP is working correctly because devices are successfully obtaining IP addresses.

DNS: Its role is to translate human-readable domain names (e.g., www.google.com) into machine-readable IP addresses. Since devices can't resolve external web addresses, the DNS configuration is failing. This happens either because the DHCP server is distributing an incorrect/unreachable DNS

server IP, firewall rules are blocking outbound DNS queries (UDP port 53), or the configured external DNS provider is experiencing an outage.

3. Explain the necessity of Inter-VLAN Routing and the specific role of static routes when a startup requires communication between two isolated departments on the same physical switch.

Answer: By default, VLANs (Virtual Local Area Networks) segment a physical switch into distinct, isolated logical networks to improve security and reduce broadcast domains. Devices on different VLANs cannot communicate directly at Layer 2, making Inter-VLAN Routing a strict necessity to allow packets to travel between them via a Layer 3 device (like a router or a Layer 3 switch).

The specific role of static routes in this context ensures explicit path mapping. While a local Layer 3 switch or router automatically builds routes for directly connected VLAN interfaces, static routes are explicitly configured by an administrator to direct traffic intended for a specific destination network through a designated next-hop gateway. In a startup network, static routes provide simple, low-overhead routing security and predictability between departmental networks without the processing requirements of dynamic routing protocols.

4. Explain the performance differences between ICMP and HTTP/HTTPS protocols that would cause a website to experience high latency while a ping test returns a fast, successful response.

Answer: This discrepancy occurs because ICMP and HTTP/HTTPS function differently across the OSI model layers and experience different processing overheads:

ICMP (Ping): Operates directly at the Network Layer (Layer 3). It requires minimal processing by network hardware and web servers, as it simply asks the target system to echo back a small packet. A fast ping means basic network-layer path connectivity is highly responsive.

HTTP/HTTPS: Operates at the Application Layer (Layer 7) over TCP (Layer 4). To load an HTTP/HTTPS website, a device must complete a 3-way TCP handshake, and for HTTPS, an additional TLS/SSL security handshake. Afterwards, the web server must process application code, query databases, and return large HTML/assets. High latency here points to application-layer or server-side issues

(e.g., high CPU usage, slow database queries, or misconfigured web servers) rather than network link problems.

5. Explain the conceptual logic of Firewall Access Control Lists (ACLs) and the strategic placement of a firewall to restrict unauthorized traffic between a startup's management and general staff departments.

Answer: Firewall Access Control Lists (ACLs) use sequential rule-based logic to evaluate incoming and outgoing packets. Each packet is inspected based on attributes such as Source IP, Destination IP, Protocol type, and Source/Destination Port numbers. The firewall evaluates the packet against the ACL rules from top to bottom; once a match is found, it executes the defined action (PERMIT or DENY) and stops processing further rules. A default 'deny all' rule typically finishes the list.

Strategic Placement: To restrict traffic between internal departments (Management and General Staff), the firewall must be placed at the boundary where these two distinct networks intersect—typically inline between the departmental VLAN gateways or directly on the inter-VLAN routing interface. This ensures that all internal traffic moving between departments must cross the firewall checkpoints, preventing unauthorized staff access to secure management segments.

Section B

Task 1: Design a VLSM subnet scheme for 3 departments requiring 15 hosts each using 192.168.10.0/24.

Answer: To support 15 hosts per department, we need to calculate the number of host bits required using the formula $2^n - 2 \geq 15$ hosts. For $n=4$, $2^4 - 2 = 14$ (not enough). For $n=5$, $2^5 - 2 = 30$ hosts, which easily fits 15. A 5 host-bit structure leaves $32 - 5 = 27$ bits for the network mask. Thus, we will use a /27 subnet mask (255.255.255.224), which increments by 32 in the fourth octet.

Departme nt	Network Address	Subnet Mask	Usable IP Range	Broadcast Address
Dept 1 (VLAN 10)	192.168.10.0/27	255.255.255.24	192.168.10.1 -	192.168.10.31

			192.168.10.3 0	
Dept 2 (VLAN 20)	192.168.10.32/ 27	255.255.255.2 24	192.168.10.3 3 - 192.168.10.6 2	192.168.10.6 3
Dept 3 (VLAN 30)	192.168.10.64/ 27	255.255.255.2 24	192.168.10.6 5 - 192.168.10.9 4	192.168.10.9 5

Task 2: Configure VLANs 10, 20, and 30 on a Cisco switch and assign physical interfaces to each.

Answer: Cisco Switch CLI Configuration script:

```
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Management
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Engineering
Switch(config-vlan)# vlan 30
Switch(config-vlan)# name Marketing
Switch(config-vlan)# exit
```

Proxy Assigning Interfaces:

```
Switch(config)# interface range fastEthernet 0/1 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fastEthernet 0/6 - 10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fastEthernet 0/11 - 15
Switch(config-if-range)# switchport mode access
```

```
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# end
Switch# write memory
```

Task 3: Set up a DHCP pool on a Router/L3 switch to provide dynamic addressing for the VLAN 10 subnet.

Answer: Cisco IOS Router Configuration script for VLAN 10 DHCP pool:

```
Router# configure terminal
Router(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.5
Router(config)# ip dhcp pool VLAN10_MGNT
Router(dhcp-config)# network 192.168.10.0 255.255.255.224
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 8.8.8.8
Router(dhcp-config)# end
Router# write memory
```

Task 4: Capture a network trace in Wireshark and identify the IP addresses in the DNS query and HTTP GET headers.

Answer: Steps to verify and identify headers within Wireshark:

Apply a display filter 'dns' or 'http' in the filter box to sort out unwanted traffic.

For DNS Query: Locate a packet labeled 'Standard query 0x... A www.example.com'. Expand the 'Domain Name System (query)' section in the details pane to view the target domain name. The source/destination IP addresses will be displayed under the Network Layer header (Internet Protocol Version 4).

For HTTP GET: Locate a packet labeled 'GET / HTTP/1.1'. Expand the 'Hypertext Transfer Protocol' section in the frame details pane to view the 'Host:' header string value. The client IP is the packet source, and the web server IP is the packet destination.

Section C

Scenario: TechLaunch, a startup with 40 employees, is experiencing "flapping" network connectivity. Employees in the Marketing wing report that their

internet drops for 30 seconds every few minutes. The Engineering wing, located on a different VLAN, remains perfectly stable. A ping test from a Marketing PC to the Core Switch shows a consistent 25% packet loss and fluctuating latency.

1. Based on the symptoms, identify the two most likely OSI layers where this fault is occurring.

Answer: The two most likely layers are:

Layer 2 (Data Link Layer): The 30-second dropping pattern occurring 'every few minutes' strongly mirrors Spanning Tree Protocol (STP) listening/learning transition states triggered by network loops or an unstable interface flapping.

Layer 1 (Physical Layer): Faulty Ethernet cabling, loose connection jacks on the Marketing patch panel, or localized electromagnetic interference can cause physical packet drops, errors, and physical link drops.

2. Outline a 4-step "Bottom-Up" troubleshooting sequence to isolate this specific issue.

Answer: Bottom-Up Troubleshooting steps:

Step 1 (Layer 1): Inspect the physical connections of the affected Marketing workstations and the specific switch ports. Check if patch cables are damaged, properly seated, or if link lights go completely amber/dark when the connection drops.

Step 2 (Layer 2): Log into the switch console to look for interface error counters (CRC errors, input errors) and view STP logs via 'show spanning-tree detail' to see if topology change notifications (TCNs) match the timeline of the drops.

Step 3 (Layer 3): Run concurrent persistent ping tests from a Marketing PC to its default gateway and check ARP tables to verify local IP resolution correctness and isolate where packets are dropped.

Step 4 (Layer 4/7): Use a network monitor analyzer tool to inspect endpoint applications if lower tiers show clear structural stability.

3. List three CLI commands you would run on the switch to check for cable faults or broadcast storms.

Answer: The three switch CLI commands are:

1. show interfaces status (Checks link operational state, speeds, and duplex settings).

2. show interfaces fastEthernet 0/x (Replace x with interface number; checks for high CRC errors or incoming runts/giants indicating bad cabling).

3. show interfaces counters storm-control (Verifies if broadcast, multicast, or unicast traffic is hitting configured storm suppression limits).

4. If the issue is a "Switching Loop," explain which protocol should have prevented this and why it might have failed.

Answer: Spanning Tree Protocol (STP / IEEE 802.1D or 802.1w) is designed to dynamically prevent switching loops by blocking redundant path ports. It might have failed here due to:

STP being manually disabled on a port or specific VLAN by an engineer.

Unmanaged switches being connected downstream by end-users in the Marketing wing, which loop back traffic internally without propagating Bridge Protocol Data Units (BPDUs).

High volume unidirectional links or delayed BPDU frames causing a switch to think a redundant link is safe to open.

5. Propose a long-term configuration change to ensure an error in one department doesn't impact the rest of the office.

Answer: Implement the following robust structural changes:

Enable BPDU Guard and Loop Guard on all edge ports configured with PortFast. This immediately disables any port if a rogue switch or loop is introduced.

Enforce Storm Control limits on all access switch ports to suppress excessive broadcast/multicast bandwidth spikes before they flood the Core Switch.

Strictly isolate user segments into non-overlapping isolated subnets with ACL limits at the Layer 3 boundaries to protect inter-departmental operations.